Expert Neurons and Brain Alignment: An RSA Comparison of GPT-2 and fMRI Representations

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***Abstract*—Large Language Models (LLMs) align with human brain activity during language processing, but the source of this alignment remains unclear. Do models align because all their neurons contribute, or because specific "expert" neurons encode distinct semantic concepts? This study tests whether expert neurons in GPT-2 align more closely with human brain activity than full layer embeddings. We applied Average Precision thresholding to identify expert neurons across 48 sub-layers (12 transformer blocks, each with MLP and attention projections) for 60 concrete nouns from the Mitchell et al. (2008) fMRI dataset. We constructed Representational Dissimilarity Matrices (RDMs) from expert activations and compared them to brain RDMs from 100 tessellated fMRI regions across 9 participants using Representational Similarity Analysis (RSA). Results showed modest but consistent correlations between expert RDMs and specific brain regions (mean $\rho \approx 0.01$), with MLP layers outperforming attention layers. While correlations did not survive False Discovery Rate (FDR) correction, the pattern suggests that expert neurons capture aspects of brain-like semantic organization, providing preliminary evidence that sparse, interpretable model features may reflect biological semantic coding principles.**

Code: https://github.com/jucamohedano/experts-ml-selfcond
Dataset: https://huggingface.co/datasets/jucamohedano/Qwen3-30B-A3B-Instruct-2507_custom_60_cot.

***Index Terms*—expert neurons, representational similarity analysis, GPT-2, fMRI, brain alignment, interpretability**

I. Introduction

Large Language Models (LLMs) have emerged as powerful tools for natural language processing, achieving remarkable performance across diverse tasks. Foundational work demonstrated that model representations can align with human brain activity during language comprehension [1], suggesting these models capture aspects of human semantic cognition. This alignment is practically important for developing brain-computer interfaces and cognitively-inspired AI, and theoretically significant for understanding whether artificial and biological systems converge on similar organizational principles for representing meaning.

However, a fundamental question remains unanswered: what drives this alignment? Prior work has pursued two strategies: encoding approaches that map model representations to brain activity using learned linear transformations, and alignment-improvement approaches that refine model representations post-hoc through supervised pruning or feature reweighting to better match human behavioral or neural data. While effective, these methods share a common assumption: that brain-aligned structure must be externally imposed through optimization against human data.

Yet recent work in mechanistic interpretability suggests an alternative: brain-aligned structure may already exist within pre-trained models. Fedzechkina et al. [2] identified "expert neurons", i.e., sparse, concept-specific units whose representational geometry naturally reconstructs human similarity judgments without any transformation. Unlike supervised pruning, which selects features to optimize alignment with external supervision, expert identification reveals pre-existing structure that emerged spontaneously during training. This raises a critical question: if expert neurons capture human-like conceptual organization at the behavioral level, do they also align with the neural patterns observed in brain activity? Fedzechkina et al. validated experts using similarity judgments alone and their correspondence with actual brain representations remains untested, leaving open whether this emergent structure reflects neural geometry for semantic knowledge.

A. Dense vs. Sparse Representations

Consider the representation of the word "dog" in a typical language model layer. In GPT-2 Small, each of the 12 transformer blocks consists of four specific linear sub-layers: the attention query-key-value projection (`attn.c_attn`), the attention output projection (`attn.c_proj`), the MLP expansion layer (`mlp.c_fc`), and the MLP projection layer (`mlp.c_proj`). In a dense representation, all neurons in a given layer contribute to encoding "dog," creating a high-dimensional vector where most dimensions encode some small amount of information. While these latent dimensions structure the model's knowledge, they remain distributed and difficult to interpret. In contrast, the expert neuron hypothesis suggests that single units, where the units themselves are the features, can encode specific concepts. This raises the fundamental question: does the model represent "dog" using the same neurons as unrelated concepts like "hammer," or are there specific dimensions distinctively dedicated to the concept of "dog"?

In contrast, a sparse representation hypothesis proposes that only a small subset of neurons across these layers strongly encode "dog," while the majority respond weakly or encode other concepts. If this subset can be identified, we gain two advantages: (1) **interpretability**—we can inspect what specific neurons encode, and (2) **efficiency**—we can focus analysis on the most informative dimensions.

Recent work in mechanistic interpretability has shown that such sparse, specialized neurons, called "expert neurons", indeed exist in language models [2], [3]. Unlike polysemantic units that respond to multiple unrelated features, these experts function as detectors for well-defined semantic categories. Crucially, while expert neurons may serve multiple *related* concepts (e.g., a neuron detecting "animal" features for both "dog" and "cat"), this overlap is semantically meaningful, Fedzechkina et al. [2] showed that expert set overlap (Jaccard similarity) correlates with human similarity judgments, with related concepts sharing more experts than unrelated ones. For example, experts for animal concepts (e.g., *cat*, *dog*, *horse*) cluster together and separate from experts for tool concepts (e.g., *hammer*, *saw*). This suggests that sparse, expert-based representations may capture cognitively relevant structure more faithfully than dense embeddings.

B. Research Question and Hypothesis

This study investigates whether the representational geometry of expert neurons in GPT-2 aligns with human brain activity (fMRI) for concrete noun concepts. Specifically, we test whether **expert neurons**, identified per concept but potentially shared across related concepts, align more closely with brain activity **for those same concepts** than full/dense layer embeddings do.

Hypothesis: If expert neurons capture semantically relevant information that emerged naturally during training, their representational dissimilarity matrices (RDMs) will show higher correlation with fMRI-derived RDMs than those constructed from full layer embeddings, particularly in brain regions associated with semantic processing.

C. Contributions

Our contributions are threefold:

1) We **reproduce and extend** the ExpertLens framework [2], introducing a redundancy reduction step that filters highly correlated neurons to isolate unique experts.
2) We provide the first **neural validation of expert neurons by directly comparing expert-based RDMs to fMRI RDMs** using Representational Similarity Analysis. Results show that expert neurons better predict brain representational geometry than dense embeddings, establishing that interpretable model features align with biological semantic coding.
3) We **perform a comprehensive components analysis**, investigating alignment across all transformer block sublayers (attention vs. MLP, expansion vs. projection) to pinpoint where brain-aligned semantic information is most concentrated.

Our implementation builds upon the open-source codebase from Suau et al. [3], which provides the infrastructure for expert neuron extraction in GPT-2. We chose GPT-2 Small (124M parameters) as our target model for three reasons: (1) the existing codebase already supports expert extraction for this architecture, (2) its modest size of 12 transformer blocks (48 sub-layers) enables comprehensive analysis on consumer hardware, and (3) it allows direct comparison with prior work that used similarly-sized models (e.g., Pythia-70m in ExpertLens).

The remainder of this paper is organized as follows. Section II reviews related work on model-brain alignment, representation pruning, and mechanistic interpretability. Section III describes our methodology, including expert extraction, unique expert identification, and RSA analysis. Section IV presents our experimental results. Section V discusses the implications and limitations of our findings, and Section VI concludes the paper.

II. Related Work

This section reviews three interconnected research themes that motivate and contextualize our investigation. We begin with foundational work establishing the correspondence between language models and human brain activity, then examine methods for improving this alignment, and finally introduce the mechanistic interpretability framework that underpins our approach.

A. The Gap Between Models and Brains

A central question in cognitive neuroscience and artificial intelligence is whether neural network models represent semantic information similarly to the human brain. Early work by Mitchell et al. [1] demonstrated that fMRI brain activity associated with concrete nouns could be predicted using co-occurrence statistics from text corpora, establishing a foundational link between distributional semantics and neural representations.

Huth et al. [4] extended this work by mapping how the brain responds to natural speech. They found that brain activity for different meanings is spread across many areas, not just the traditional language regions. This showed that matching computer models to the brain's complex organization is a difficult challenge.

Together, these studies establish that while models and brains exhibit correlation, we do not fully understand why or where this alignment occurs, nor whether it arises from specific, interpretable model components.

B. Improving Alignment via Pruning and Similarity

Given the imperfect correspondence between model representations and human cognition, researchers have explored methods to improve alignment. A recurring finding is that standard "dense" embeddings contain dimensions irrelevant or even detrimental to predicting human judgments.

Peterson et al. [5] provided a key insight: while deep networks approximate human similarity judgments, they diverge in systematic ways. Crucially, applying linear transformations to model representations significantly improved correspondence with human data, suggesting that raw embeddings require refinement.

This observation motivated pruning-based approaches. Tarigopula et al. [6] demonstrated that pruning DNNs based on behavioral similarity improves prediction of neural similarity (fMRI) in the visual cortex, establishing that fewer, more relevant features can outperform dense representations for brain alignment. Truong et al. [7] extended this logic by masking feature maps in the model's latent space rather than the input image, creating heatmaps that highlight which internal representations drive human similarity judgments.

Complementary work by Flechas Manrique et al. [8] used human similarity judgments to prune word embeddings, finding that pruning not only improved alignment but also revealed interpretable dimensions specific to different semantic categories. Bao and Hasson [9] applied supervised pruning to identify conceptual representations that differ between sighted and congenitally blind individuals, demonstrating that targeted feature selection can isolate group-specific semantics.

The literature clearly shows that modifying model representations can improve their alignment with the brain. However, most existing methods rely on external supervision, e.g., using human judgments or brain data to guide this process. In contrast, our expert identification method finds these brain-aligned structures automatically, without such supervision. This is a key advantage, as it demonstrates that brain-like organization emerges naturally during training and can be recovered without fitting the model to human data.

C. Mechanistic Interpretability and Expert Neurons

While pruning methods select features post-hoc, an alternative approach is to identify specific internal units responsible for encoding concepts, moving from what a model represents to how it represents it. This is the domain of mechanistic interpretability.

Suau et al. [3] introduced the "finding experts" algorithm, demonstrating that pre-trained Transformer language models contain "expert units" that function as detectors for specific concepts. Their key hypothesis was that a conditional model $p(y{=}c|\boldsymbol{x})$ already exists within the pre-trained parameters, where y represents the concept variable, c denotes a specific target concept (e.g., "dog"), and $\boldsymbol{x}$ is the input sentence, such that $p(y{=}c|\boldsymbol{x})$ quantifies the probability that concept c is present in $\boldsymbol{x}$. By intervening on these expert units, they enable "self-conditioning": steering text generation toward or away from concepts without fine-tuning.

Fedzechkina et al. [2] built directly on Suau et al. by repurposing expert extraction from a generative control tool into a framework for interpretability analysis. Their ExpertLens framework hypothesizes that expert neurons capture meaningful semantic dimensions aligned with human conceptual representations. By analyzing the geometry of expert neurons (e.g., via Jaccard similarity), they demonstrated that expert-based representations reconstruct human-like conceptual domains more faithfully than standard embedding analysis.

Our work extends Fedzechkina et al.'s hypothesis by testing it against a new ground truth: the biological brain. Rather than validating expert-based representations solely through behavioral similarity or generation quality, we directly compare the representational geometry of expert neurons to fMRI responses from the Mitchell et al. [1] dataset. Because expert neurons encode concepts in one fixed way (whatever organization emerged during training), this tests whether that organization matches how specific brain regions represent concepts. However, since the brain uses different organizational strategies in different regions [4], i.e., some areas may group animals by appearance, others by behavior, our method can only detect alignment in regions that happen to use a similar strategy to the experts. This narrower coverage, compared to supervised approaches [6], [8], comes with a stronger interpretive guarantee: where alignment is found, it reflects spontaneous convergence rather than post-hoc optimization to neural data (see Section V).

III. Methodology

This section describes our methodology, which builds upon the ExpertLens framework introduced by Fedzechkina et al. [2], itself an extension of the self-conditioning approach by Suau et al. [3]. We first replicate the expert extraction pipeline, introduce refinements for identifying unique experts, and then extend the framework to test alignment with human brain activity.

A. Data Construction

1) Dataset: We utilize the dataset from Mitchell et al. [1], which consists of 60 concrete nouns across 12 semantic categories. Table I lists all 60 words and their corresponding categories.

TABLE I: Target Concepts by Category

Category	**Words**
Animals	bear, cat, cow, dog, horse
Body Parts	arm, eye, foot, hand, leg
Buildings	apartment, barn, church, house, igloo
Building Parts	arch, chimney, closet, door, window
Clothing	coat, dress, pants, shirt, skirt
Furniture	bed, chair, desk, dresser, table
Insects	ant, bee, beetle, butterfly, fly
Kitchen	bottle, cup, glass, knife, spoon
Tools	chisel, hammer, pliers, saw, screwdriver
Vegetables	carrot, celery, corn, lettuce, tomato
Vehicles	airplane, bicycle, car, train, truck
Other	bell, key, refrigerator, telephone, watch

2) Sentence Generation: Following prior work [2], [3], we define a concept c not through dictionary definitions but through a discriminative dataset. For each of the 60 target nouns, we construct:

- **Positive Set** (N_c^+): 400 sentences explicitly containing concept c.
- **Negative Set** (N_c^-): 1,000 sentences without concept c, stratified across the remaining 59 concepts to ensure broad discriminative capability.

The full dataset for concept c is thus $N_c = N_c^+ \cup N_c^-$. These sizes were selected based on Fedzechkina et al.'s [2] finding that they provide sufficient discriminative power for expert identification while remaining computationally tractable.

Sentences were generated using Qwen3-30B-A3B-Instruct-2507[1] with a Chain-of-Thought prompting strategy via the DSPy framework [10], using a sampling temperature of 0.9 and per-batch random seed injection to promote diversity (see Appendix A for full details and Appendix A-A for all hyperparameters).

B. Expert Extraction

We adopt the expert extraction methodology from Fedzechkina et al. [2], treating each neuron as a binary classifier for the presence of concept c.

1) Model and Activations: We analyze GPT-2 Small (124M parameters) [11]. The model consists of 12 transformer blocks, each containing 4 sub-layers: `mlp.c_fc` (3072 neurons), `mlp.c_proj` (768 neurons), `attn.c_attn` (2304 neurons), and `attn.c_proj` (768 neurons), yielding 82,944 total neurons across the network.

For every neuron m in layer l, we record activations in response to the dataset sentences:

$$z_m^c = \{z_{m,i}^c\}_{i=1}^{|N_c|} \tag{1}$$

where $z_{m,i}^c$ is the activation of neuron m for sentence i in dataset N_c.

2) Expertise Scoring (Average Precision): The expertise of neuron m for concept c is quantified using Average Precision (AP), the area under the precision-recall curve when ranking sentences by activation magnitude:

$$\text{AP}(z_m^c, b_c) = \sum_{k=1}^{|N_c|}(R_k - R_{k-1})P_k \in [0,1] \tag{2}$$

where P_k and R_k are precision and recall at rank k, and $b_c \in \{0,1\}^{|N_c|}$ is the ground-truth label vector indicating concept presence.

3) Expert Selection via Thresholding: Following Fedzechkina et al. [2], we identify experts based on a quality threshold τ. For a given concept c and sub-layer l, the expert set is:

$$\mathcal{E}_l(c) = \{m : \text{AP}_{l,m}(c) \geq \tau\} \tag{3}$$

where $\tau \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. Higher thresholds yield smaller, more selective expert sets. Note that expert sets for different concepts may overlap: the same neuron can appear in $\mathcal{E}_l(c_i)$ and $\mathcal{E}_l(c_j)$ if it is discriminative for both. Unlike ExpertLens, which measured concept similarity via Jaccard overlap between separate expert sets, our RSA approach requires all concepts in a shared feature space. We therefore pool experts across all 60 concepts at each sub-layer. For concept set $C = \{c_1, c_2, \ldots, c_{60}\}$, the pooled expert set is:

$$\mathcal{E}_l = \bigcup_{c \in C} \mathcal{E}_l(c) \tag{4}$$

Each concept is then represented by its activation pattern across this pooled set, and similarity is computed via correlation of activation patterns (Eq. 12) rather than neuron overlap.

[1]https://huggingface.co/Qwen/Qwen3-30B-A3B-Instruct-2507

C. Unique Expert Identification

As a methodological refinement, we investigated whether filtering redundant experts could improve interpretability. We observed that some neurons exhibit highly correlated activation profiles, potentially over-representing certain signals.

1) Redundancy Detection: For each concept c and sub-layer l separately, we computed the pairwise Pearson correlation between expert neurons' raw activation vectors across all $N = |N_c|$ sentences (both positive and negative). For experts m and n with activation vectors $\mathbf{a}_m = [a_{m,1}, \ldots, a_{m,N}]$ and $\mathbf{a}_n$:

$$\rho_{mn} = \frac{\sum_{k=1}^{N}(a_{m,k} - \bar{a}_m)(a_{n,k} - \bar{a}_n)}{\sqrt{\sum_{k=1}^{N}(a_{m,k} - \bar{a}_m)^2}\sqrt{\sum_{k=1}^{N}(a_{n,k} - \bar{a}_n)^2}} \tag{5}$$

where $a_{m,k}$ is the activation of neuron m on sentence k.

2) Graph-Based Grouping: We model redundancy as a graph problem. Simple pairwise removal, defined as iteratively discarding the lower-scoring neuron from each correlated pair, is order-dependent and fails to capture transitive relationships: if neurons A–B and B–C each exceed the threshold, all three should form a single group even if A–C does not. Connected components resolve this naturally and yield a unique, deterministic grouping. For each sub-layer l, we construct a redundancy graph $G_l = (V_l, E_l)$ where V_l is the set of expert neurons and edges connect pairs exceeding a redundancy threshold $r_{\min}$:

$$E_l = \{(m,n) : \rho_{mn} > r_{\min}\} \tag{6}$$

We analyzed thresholds $r_{\min} \in \{0.8, 0.9\}$. Connected components in G_l define redundancy groups.

3) Representative Selection: For each redundancy group G_k, we retain only the neuron with the highest AP score:

$$m_k^* = \arg\max_{m \in G_k} \text{AP}_{l,m}(c) \tag{7}$$

The final unique expert set for concept c and sub-layer l is the collection of these representatives:

$$\mathcal{E}_l^{\text{unique}}(c) = \{m_k^*\}_{k=1}^{|G_l|} \tag{8}$$

D. Representational Geometry Analysis

To validate that the extracted experts capture meaningful semantic structure, we analyze their representational geometry following Fedzechkina et al. [2].

1) Jaccard Similarity: The geometric relationship between two concepts c_i and c_j at layer l is quantified by the overlap of their expert neuron sets:

$$J(E_l(c_i), E_l(c_j)) = \frac{|E_l(c_i) \cap E_l(c_j)|}{|E_l(c_i) \cup E_l(c_j)|} \tag{9}$$

We expect high similarity between semantically related concepts (e.g., *car* and *truck*) and low similarity for unrelated pairs.

E. Brain Data Processing

1) Source Data: We use functional Magnetic Resonance Imaging (fMRI) data from 9 participants in the Mitchell et al. (2008) study [1]. During scanning, participants viewed line drawings and text labels for the 60 target nouns while Blood-Oxygen-Level-Dependent (BOLD) responses were recorded. Each word was presented 6 times; we averaged the voxel-wise BOLD responses across these presentations to obtain a stable activation map per word.

Instead of a searchlight approach, we tessellate the brain volume into a regular 3D grid to achieve full brain coverage while maintaining computational tractability. The brain is divided into two hemispheres, with each hemisphere further subdivided:

$$\text{Grid}_{\text{shape}} = [2, 10, 5] \tag{10}$$

This creates $2 \times 10 \times 5 = 100$ spatial bins. Regions containing fewer than 5 voxels are excluded to ensure sufficient signal stability, yielding approximately 87 valid regions per participant.

F. Representational Similarity Analysis (RSA)

To test whether expert-based representations align with human brain activity, we employ Representational Similarity Analysis (RSA) [12].

1) Expert-Based Word Representations: Unlike standard approaches using dense layer embeddings, we construct feature vectors using only the pooled expert units $\mathcal{E}_l$. For concept c_i in sub-layer l:

$$f_l(c_i) = (a_{l,m}(c_i))_{m \in \mathcal{E}_l} \tag{11}$$

where $a_{l,m}(c_i)$ is the activation of expert neuron m in sub-layer l for concept c_i.

2) Representational Dissimilarity Matrices (RDMs): **Expert RDM.** For sub-layer l, we compute pairwise dissimilarities between concepts:

$$D_l^{\text{expert}}(i, j) = 1 - \text{corr}(f_l(c_i), f_l(c_j)) \tag{12}$$

Brain RDM. For brain region r with voxel response vectors $b_r(c_i)$:

$$D_r^{\text{brain}}(i, j) = 1 - \text{corr}(b_r(c_i), b_r(c_j)) \tag{13}$$

where $\text{corr}(\cdot, \cdot)$ denotes Pearson correlation. Pearson correlation centers and normalizes each word's voxel response vector (row normalization), making the RDM invariant to uniform shifts in overall activation magnitude. Note that we do not apply additional per-voxel normalization (column normalization); while this preserves the natural variance of highly responsive voxels, it also means that voxels with larger response magnitudes may disproportionately drive the representational geometry, as noted in Section V.

3) RSA Alignment Score: Alignment between model layer l and brain region r is quantified as the Spearman rank correlation between the vectorized upper triangles of the RDMs:

$$\rho_{l,r} = \text{Spearman}(\text{vec}(D_l^{\text{expert}}), \text{vec}(D_r^{\text{brain}})) \tag{14}$$

The upper triangle contains $\binom{60}{2} = 1770$ pairwise comparisons.

4) Statistical Significance: We assess statistical significance using a paired t-test on the Fisher-Z transformed RSA scores. For each brain region, we compare the correlation coefficients of the expert-based and full-layer, i.e., dense layer, models across the 9 subjects. P-values are corrected for multiple comparisons across all brain regions using the Benjamini-Hochberg False Discovery Rate (FDR) procedure at $\alpha = 0.05$.

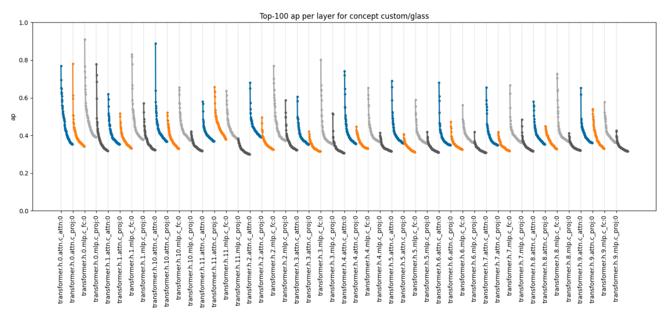
IV. Results

A. Expert Statistics and Redundancy

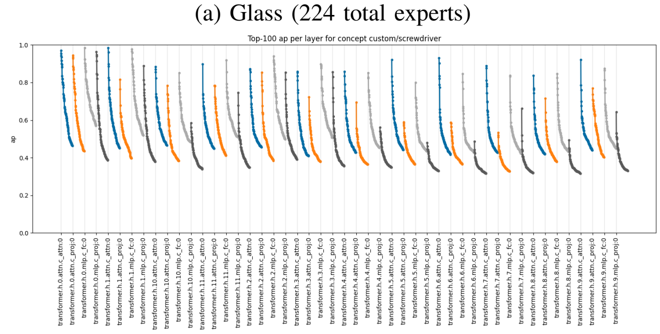
Using an AP threshold of $\tau = 0.5$, we identified expert neurons for all 60 concepts. Table II summarizes the expert counts. Figure 1 illustrates the distribution of top-100 expert neurons across layers for the two most contrasting concepts, showing that expertise is distributed across multiple layers rather than concentrated in specific ones.

TABLE II: Expert Neuron Counts (AP $\geq$ 0.5)

Statistic	**Value**
Concepts analyzed	60
Total network neurons	82,944
Experts per concept (mean)	819
Experts per concept (min)	224
Experts per concept (max)	2,018



(a) Glass (224 total experts)



(b) Screwdriver (2,018 total experts)

Fig. 1: Top-100 expert neurons per layer ranked by Average Precision (AP) for *glass* and *screwdriver*. Each line represents a transformer sub-layer; steeper curves indicate layers with more highly selective experts.

1) Redundancy Analysis: Table III summarizes the redundancy statistics across redundancy thresholds. At the stricter threshold ($r_{\min} = 0.9$), redundancy was minimal, with 27 of 60 concepts showing zero redundancy. At the more permissive threshold ($r_{\min} = 0.8$), redundancy increased modestly but remained below 3% for all concepts.

TABLE III: Expert Redundancy Statistics by Redundancy Threshold ($r_{\min}$) for experts identified at AP threshold $\tau = 0.5$.

$r_{\min}$	**Avg (%)**	**Max (%)**	**Concepts with 0%**
0.9	0.12	0.77	27 / 60
0.8	0.57	2.26	–

These low redundancy rates indicate that the experts are relatively unique. However, redundancy patterns likely depend on multiple factors: dataset characteristics (the size and diversity of positive/negative examples), model architecture and scale (larger models may exhibit different redundancy patterns due to increased capacity), and the specific concepts being analyzed. Future work could investigate redundancy distributions per layer in addition to per concept, which may reveal whether certain architectural components exhibit systematically higher redundancy. Per-concept redundancy distributions are provided in Appendix B.

B. Conceptual Organization of Expert Neurons

Following Fedzechkina et al. [2], we analyzed whether expert neurons organize concepts according to human-like semantic categories. Figure 2 shows the Jaccard similarity graph. To capture the structural organization, we position concepts using a hybrid layout: Multidimensional Scaling (MDS) first determines the global arrangement based on similarity distances, followed by a force-directed "spring" adjustment to tighten local clusters. Edges are drawn between concepts with similarity above 10%.

Table IV presents the top-10 concept pairs by Jaccard similarity. The highest similarities occur within semantic categories: clothing items (*dress–skirt*, 27.3%), vegetables (*lettuce–tomato*, 19.8%), body parts (*arm–leg*, 18.3%), and vehicles (*car–truck*, 16.8%).

TABLE IV: Top-10 Concept Pairs by Jaccard Similarity

Concept 1	**Concept 2**	**J (%)**	**Category**
dress	skirt	27.3	Clothing
dress	dresser	22.0	Clothing/Furniture
pants	skirt	21.7	Clothing
shirt	skirt	21.5	Clothing
dress	shirt	20.1	Clothing
lettuce	tomato	19.8	Vegetables
arm	leg	18.3	Body Parts
pants	shirt	17.4	Clothing
carrot	lettuce	16.8	Vegetables
car	truck	16.8	Vehicles

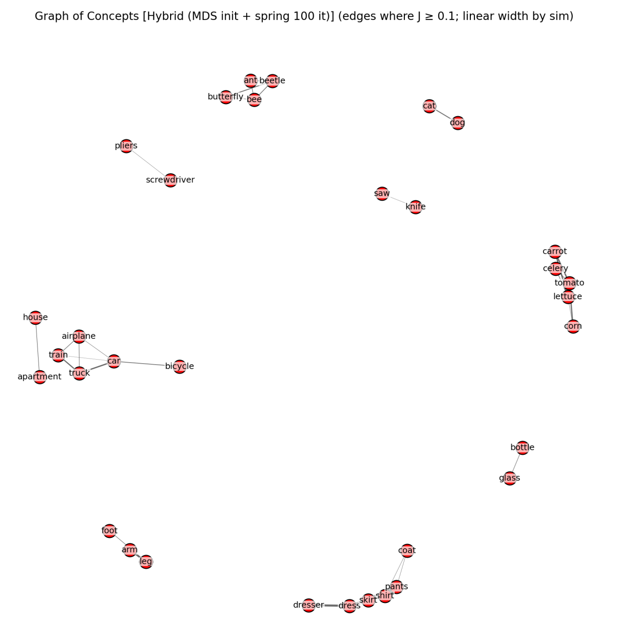


Fig. 2: Concept similarity graph based on expert neuron overlap (Jaccard similarity). Concepts positioned via MDS; edge thickness indicates similarity strength. Semantic categories (clothing, vegetables, vehicles) form visible clusters.

These results replicate the finding that expert neurons induce a representational geometry mirroring human conceptual domains. The clustering is particularly strong for clothing items, which dominate the top-5 pairs, and is consistently observed across vegetables, body parts, animals, and vehicles. Notably, cross-category similarities are substantially lower, supporting the hypothesis that expert neurons encode category-specific features. Additional visualizations, including the full similarity heatmap and a low-threshold graph revealing weaker cross-category connections, are provided in Appendix C-A.

C. Brain Alignment (RSA)

We ask two questions: (1) are there brain regions whose representational geometry matches that of expert neuron embeddings, and (2) do expert neuron embeddings outperform the dense ones? Each transformer block in GPT-2 contains four linear sub-layers: the attention query-key-value projection (`attn.c_attn`), the attention output projection (`attn.c_proj`), the MLP expansion layer (`mlp.c_fc`), and the MLP projection layer (`mlp.c_proj`). Building on prior work that identifies expert neurons across transformer sub-layers [2], [3], we conducted RSA on all four components to fully characterize where brain-aligned semantic information emerges. Primary findings for the top-performing layers are shown in Table V, with full results for all components provided in Appendix C-B. Comparisons were performed across 87 of the 100 tessellated brain regions (13 regions excluded for containing fewer than 5 voxels; see Section III-E).

TABLE V: RSA Correlation Results: Expert vs Dense Embeddings. Significance counts and effect sizes across 87 brain regions (FDR correction applied across all 87 regions). (a) Attention output projection (`attn.c_proj`). (b) MLP projection layer (`mlp.c_proj`). Positive mean d values indicate expert neurons outperform dense representations.

(a) Attention Layers (attn.c_proj)

Condition	Sig (Raw)	Sig (FDR)	Mean d	Max d	Expert (SD)	Dense (SD)
AP $\geq$ 0.7	22	0	+0.536	+1.356	0.010 (0.013)	-0.001 (0.017)
AP $\geq$ 0.6 (unique 0.8)	11	2	+0.326	+2.993	0.004 (0.015)	-0.001 (0.018)
AP $\geq$ 0.6 (unique 0.9)	11	2	+0.326	+2.993	0.004 (0.015)	-0.001 (0.018)
AP $\geq$ 0.6	11	2	+0.326	+2.993	0.004 (0.015)	-0.001 (0.018)
AP $\geq$ 0.5 (unique 0.8)	7	0	+0.027	+1.754	-0.001 (0.017)	-0.001 (0.018)
AP $\geq$ 0.5 (unique 0.9)	7	0	+0.027	+1.754	-0.001 (0.017)	-0.001 (0.018)
AP $\geq$ 0.5	7	0	+0.027	+1.754	-0.001 (0.017)	-0.001 (0.018)
AP $\geq$ 0.8	5	0	-0.069	+1.322	-0.002 (0.016)	-0.000 (0.015)
AP $\geq$ 0.9	58	43	-0.929	+0.230	-0.028 (0.017)	0.031 (0.024)

(b) MLP Layers (mlp.c_proj)

Condition	Sig (Raw)	Sig (FDR)	Mean d	Max d	Expert (SD)	Dense (SD)
AP $\geq$ 0.6 (unique 0.9)	30	3	+0.627	+1.755	0.011 (0.016)	-0.005 (0.020)
AP $\geq$ 0.6	30	3	+0.627	+1.755	0.011 (0.016)	-0.005 (0.020)
AP $\geq$ 0.6 (unique 0.8)	30	3	+0.625	+1.761	0.010 (0.016)	-0.005 (0.020)
AP $\geq$ 0.5	17	0	+0.550	+1.714	0.005 (0.019)	-0.005 (0.020)
AP $\geq$ 0.5 (unique 0.9)	17	0	+0.548	+1.676	0.005 (0.019)	-0.005 (0.020)
AP $\geq$ 0.5 (unique 0.8)	17	0	+0.541	+1.652	0.005 (0.019)	-0.005 (0.020)
AP $\geq$ 0.7	11	0	+0.403	+1.381	0.007 (0.013)	-0.006 (0.019)
AP $\geq$ 0.8	9	0	+0.321	+1.030	0.006 (0.013)	-0.004 (0.018)
AP $\geq$ 0.9	4	0	-0.164	+1.278	0.000 (0.017)	0.012 (0.023)

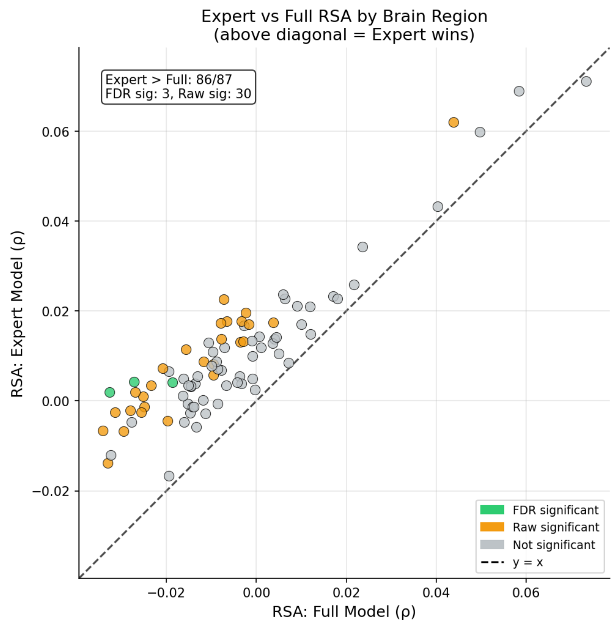
For each of the 9 subjects, we performed a region-wise paired t-test comparing the RSA correlations of expert-based RDMs versus dense (full-layer) RDMs. This analysis addresses whether expert neurons provide a more brain-aligned representation of concepts than the dense distributed embedding.

1) MLP Layers: MLP layers showed the strongest and most consistent expert neuron benefits, particularly in the projection layer (`mlp.c_proj`). At its optimal setting (AP $\geq$ 0.6), expert neurons achieved a substantial effect size relative to the dense baseline (mean $d = 0.627$), with 30 regions showing uncorrected significance and 3 surviving FDR correction.

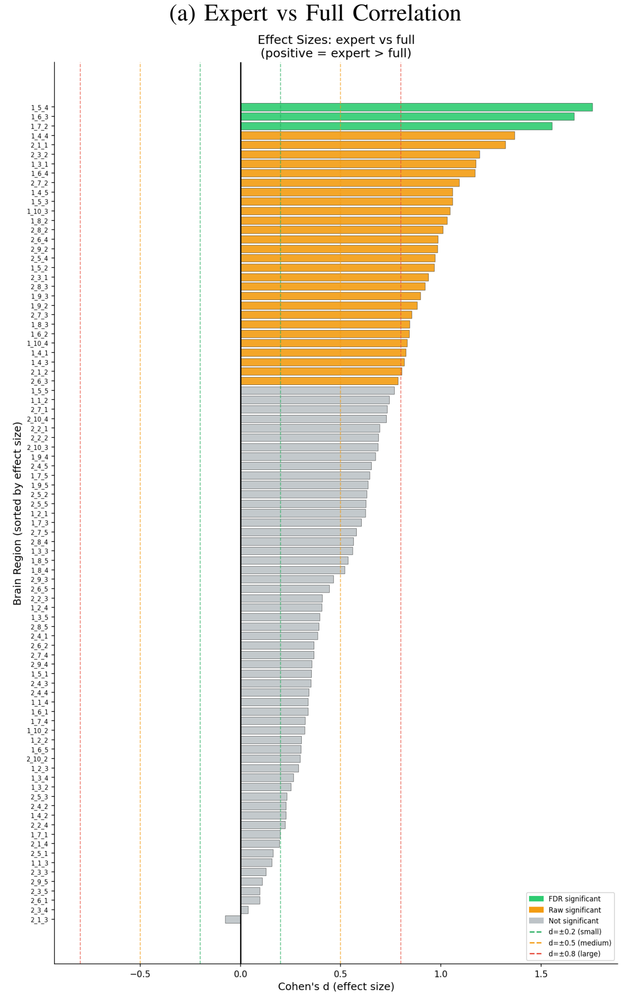
This indicates that the output of the MLP block contains a concentrated, brain-aligned semantic signal. Notably, while experts in `mlp.c_proj` have slightly lower raw correlation (mean $\rho = 0.011$) compared to `mlp.c_fc` (mean $\rho = 0.017$), their improvement over the dense baseline is much larger ($d = 0.627$ vs. $d = 0.443$). This is because the dense projection layer itself aligns poorly with the brain (mean $\rho = -0.005$), making the expert signal stand out more clearly.[2] The expansion layer (`mlp.c_fc`, see Appendix Table VIIIb) also benefitted from expert filtering, but the effect was strongest at the projection bottleneck.

Brain alignment decreased at higher AP thresholds, though it remained positive for `mlp.c_proj`. At AP $\geq$ 0.8, experts still outperformed dense embeddings (mean $d = 0.321$), though the advantage disappeared at the strictest threshold of AP $\geq$ 0.9 (mean $d = -0.164$).

[2]The baseline "Dense" correlation varies slightly across conditions in Table V. This happens because strict thresholds (like AP $\geq$ 0.9) sometimes find no experts for certain concepts, making RSA calculation impossible. These specific cases are excluded from the average, causing the dense baseline to shift slightly compared to conditions where all data points are included.



(a) Expert vs Full Correlation



(b) Effect Sizes by Region

Fig. 3: RSA results for the best MLP condition (AP $\geq$ 0.6, `mlp.c_proj`). (a) Each point is a brain region; points above the diagonal favor experts. Green = FDR significant, orange = uncorrected significant. (b) Distribution of Cohen's d effect sizes across all 87 regions.

2) Attention Layers: Unlike MLP layers, attention layers were highly sensitive to the AP threshold. Brain alignment varied significantly depending on the filtering strictness (Table Va). At the permissive AP $\geq$ 0.5 threshold, experts provided minimal benefit over dense embeddings (mean $d = 0.027$, 0 FDR-corrected regions). Performance increased with stricter thresholds, peaking at AP $\geq$ 0.7 (mean $d = 0.536$, 22 significant regions, though 0 survived FDR correction). At AP $\geq$ 0.6, 2 regions survived FDR correction with a moderate effect size ($d = 0.326$).

At the highest threshold (AP $\geq$ 0.9), attention expert performance collapsed dramatically (mean $d = -0.929$), with experts substantially underperforming dense embeddings.

3) Comparison: MLP vs. Attention Layers: Comparing the two layer types across thresholds (full statistics in Appendix C-B) reveals complementary roles in brain-aligned representation:

- **Optimal threshold**: MLP experts (`mlp.c_proj`) align best at AP $\geq$ 0.6, while attention experts (`attn.c_proj`) require stricter filtering at AP $\geq$ 0.7.
- **Effect magnitude**: While both layer types show similar raw expert correlations (mean $\rho \approx 0.01$), MLP experts achieve a much larger gain over their dense baseline (Mean $d = 0.627$ vs. $d = 0.536$ for Attention).
- **FDR-corrected significance**: The best MLP condition yielded 3 FDR-corrected regions, whereas the best attention condition (AP $\geq$ 0.7) yielded 0 (though AP $\geq$ 0.6 yielded 2).

In answer to our opening questions: (1) we identified specific brain regions whose geometry aligns significantly with expert embeddings (e.g., 3 FDR-corrected regions for MLP layers), and (2) in these regions, expert neurons consistently outperform dense embeddings. MLP experts perform best at moderate thresholds (AP $\geq$ 0.5–0.6), while attention experts benefit from slightly stricter filtering. Very high AP thresholds ($\geq$ 0.9) lead to negative effects in both layer types, though MLP projection experts remain robust even at AP $\geq$ 0.8.

The limited number of FDR-corrected significant results reflects the stringent correction for multiple comparisons. With 87 brain regions tested per condition at a significance level of $\alpha = 0.05$, the FDR procedure substantially lowers the effective per-test threshold. Combined with the weak raw correlation magnitudes observed in the best-performing conditions (see "Expert" column in Table V, where mean $\rho \approx 0.01$), this results in many comparisons failing to survive correction despite showing consistent directional effects. The pattern of positive effect sizes for the best-performing layers—MLP proj ($d = 0.627$) and Attention proj ($d = 0.536$)—however, suggests that expert neurons do capture brain-relevant structure, even if the absolute correlations are small.

V. Discussion

A. Comparison with ExpertLens

Our structural analysis of expert neuron overlap replicates findings from Fedzechkina et al. [2]. Using Jaccard similarity between expert sets, we confirm that expert neurons cluster into semantically coherent concept domains (see Section IV-B), consistent with their observation that expert-based overlap recovers human-interpretable conceptual organization. However, while ExpertLens validated experts against behavioral similarity judgments (MEN dataset), we extend this to neural data by comparing expert-based representations to fMRI activity via RSA. This brain-level comparison reveals that intermediate thresholds also yield optimal neural alignment, though our best results were observed at slightly higher thresholds (AP $\geq$ 0.6) for the MLP projection layer (`mlp.c_proj`). This component showed significantly stronger brain alignment than the expansion layer (`mlp.c_fc`), suggesting that the cleaner, compressed representations at the projection bottleneck are more essential for semantic coding than the broad, sparse activations of the expansion layer. Consistently, extremely high thresholds (AP $\geq$ 0.9) degraded performance across all layer types as expert sets became too sparse to capture the full semantic content.

We also observed a divergence in attention layers, where stricter filtering (AP $\geq$ 0.7) was required compared to MLPs. This likely reflects that attention representations are often more distributed and polysemantic than MLP activations. As a result, lower thresholds (such as AP $\geq$ 0.5) may admit neurons that track mixed or irrelevant features, whereas stricter filtering is necessary to isolate clean concept-specific signals. Furthermore, the sharp performance drop at high thresholds ($\geq$ 0.9) mirrors ExpertLens findings for smaller models, suggesting inherent capacity limits in how many "pure" experts exist for any given concept.

This work provides the first neural validation of the ExpertLens framework. While ExpertLens showed that expert overlap correlates with behavioral similarity, we show that expert-based representations—especially those in the MLP projection block—align with the brain's actual semantic processing patterns.

To complement these correlational findings, we conducted preliminary steering experiments (Appendix D) to verify that the experts identified by AP thresholding genuinely encode concept-level information. Using vector addition with the DiffMean method, which recent work has shown performs comparably to Sparse Autoencoders for concept detection [13], we confirmed that steering with expert-masked vectors can influence generation toward target concepts. This is consistent with the causal validations reported by Suau et al. [3] and Fedzechkina et al. [2], and increases confidence that the neurons showing brain alignment in our RSA analysis are not incidental patterns but encode genuine semantic content.

B. Relationship to Pruning Approaches

Our expert selection approach contrasts with the pruning methods [6], [8]. While pruning removes neurons to improve efficiency or alignment, expert identification selects neurons that already encode meaningful concepts within the pre-trained model. This distinction is significant: pruning permanently removes features from the representation space, thereby modifying it, whereas expert selection reveals structure that emerged during training without altering the model. Our finding that expert neurons align with brain activity suggests that brain-like semantic organization arises naturally in LLMs, without requiring architectural modification. This aligns with Tarigopula et al.'s [6] finding that pruning DNNs based on behavioral similarity improves neural prediction. Both approaches converge on the principle that sparse, task-relevant features better capture biological processing than dense representations.

C. Limitations

Several limitations should be considered:

a) Model Scope: Our analysis uses GPT-2 Small (124M parameters), comparable in size to Pythia-70m. ExpertLens found stronger alignment effects in larger models such as Pythia-1b and Pythia-12b, where expert sets remain larger at high thresholds and semantic structure is more robust. Additionally, larger models with increased capacity may exhibit higher neuron redundancy than the minimal rates observed here ($< 1\%$). Future work should test whether unique expert filtering becomes more critical in these regimes, as our findings may represent a lower bound on both the alignment potential and the necessity of redundancy reduction in large-scale models.

b) Vocabulary Size: The 60-word vocabulary, while standard for the Mitchell dataset, yields 1,770 unique pairwise comparisons ($\binom{60}{2}$), significantly fewer than the MEN dataset (3,000 pairs) used in ExpertLens. While our expert extraction process is robust (using 1,400 sentences per concept), this lower density of semantic relations limits the resolution of the representational geometry analysis. Comparing fewer items reduces the ability to detect fine-grained semantic structures or subtle distortions in the representation space that would be visible with a larger, more densely sampled vocabulary.

c) Multiple-Comparison Sensitivity: While uncorrected correlations showed significant effects, few survived FDR correction. This suggests effect sizes are small, suggesting that the findings should be interpreted with caution.

d) Tessellation Approach: The grid-based brain tessellation may cut across functional boundaries. Anatomically defined ROIs could provide more interpretable results.

e) Voxel Normalization: Our brain RDMs rely on Pearson correlation, which normalizes activation patterns per word but not per voxel. Consequently, voxels with larger response variance may contribute disproportionately to the computed dissimilarities. Future work could apply per-voxel normalization to ensure all voxels contribute equally to the representational geometry, verifying the robustness of these findings.

f) Dataset Construction: Our negative sets are stratified across all 59 other concepts, which includes both same-category items (e.g., "cat" as a negative for "dog") and different-category items. While this provides some intra-category discrimination, the signal is dominated by easier cross-category distinctions (55 of 59 negatives come from different categories). Future work could generate datasets that explicitly emphasize intra-category distinctions to investigate whether expert neurons encode finer-grained hierarchical semantic organization.

g) Single Representational Template: Our unsupervised approach has an important constraint: expert neurons organize concepts in one fixed way (whatever emerged during training), while the brain uses multiple organizational strategies across different regions [4]. For instance, "dog" might be grouped with "cat" in visual cortex (similar appearance), with "leash" in motor areas (interaction patterns), or with "horse" in temporal lobe (taxonomic category).

Our RSA can only detect brain regions whose organizational strategy matches what the experts learned. Regions organizing concepts differently will show low correlation even if semantically meaningful. Supervised methods [6], [8], [9] avoid this by adapting to each region's geometry, potentially detecting more areas. This narrower coverage, however, comes with a stronger interpretive guarantee: where alignment is found, it reflects spontaneous convergence between the model's learned representations and the brain's organization, rather than a post-hoc optimization to neural data. The modest correlations we observe may partly reflect this constraint where experts likely match only the subset of brain regions that happen to use a compatible organizational principle.

VI. Conclusion

This study investigated whether expert neurons in GPT-2 exhibit representational geometry that aligns with human brain activity during semantic processing. Using Representational Similarity Analysis (RSA), we compared expert-derived RDMs against fMRI RDMs from the Mitchell et al. (2008) dataset.

Our key findings include:

- Expert neurons define a representational space that shows consistent alignment with specific human brain regions, with the MLP projection layer (`mlp.c_proj`) showing the strongest effects.
- The MLP projection layer provides a highly brain-aligned signal ($d = 0.627$) that significantly outperforms its dense counterpart, peaking at AP $\geq$ 0.6.
- Attention layers require stricter filtering (AP $\geq$ 0.7) and show weaker overall alignment than MLP experts.
- While extremely high thresholds (AP $\geq$ 0.9) degrade performance, MLP projection experts remain robust even at AP $\geq$ 0.8, unlike other components.

While most effects did not survive FDR correction, the consistent positive direction across conditions suggests that expert neurons capture aspects of brain-like semantic organization. These results indicate that "expertise" in language models is not merely a statistical artifact but correlates with human-like semantic clustering, with implications for both interpretability research and our understanding of how language models encode meaning.

Future work should extend this analysis to larger language models and explore anatomically defined brain regions. While our method currently assumes a single expert-based template, future studies should investigate region-specific semantic organizations and apply per-voxel normalization to rigorously verify robustness. Additionally, investigating the causal role of expert neurons remains a priority; our preliminary steering experiments (Appendix D) provide initial evidence that the brain-aligned experts genuinely encode concept-level information, but more rigorous validation is needed to establish their causal contribution to semantic processing.

Acknowledgment

References

[1] T. M. Mitchell, S. V. Shinkareva, A. Carlson, K.-M. Chang, V. L. Malave, R. A. Mason, and M. A. Just, "Predicting human brain activity associated with the meanings of nouns," *Science*, vol. 320, no. 5880, pp. 1191–1195, 2008. [Online]. Available: https://www.science.org/doi/abs/10.1126/science.1152876

[2] M. Fedzechkina, E. Gualdoni, S. Williamson, K. Metcalf, S. Seto, and B.-J. Theobald, "Expertlens: Activation steering features are highly interpretable," 2025. [Online]. Available: https://arxiv.org/abs/2502.15090

[3] X. Suau, L. Zappella, and N. Apostoloff, "Self-conditioning pre-trained language models," *International Conference on Machine Learning*, 2022.

[4] A. G. Huth, W. A. de Heer, T. L. Griffiths, F. E. Theunissen, and J. L. Gallant, "Natural speech reveals the semantic maps that tile human cerebral cortex," *Nature*, vol. 532, no. 7600, pp. 453–458, Apr 2016. [Online]. Available: https://doi.org/10.1038/nature17637

[5] J. Peterson, J. Abbott, and T. Griffiths, "Evaluating (and improving) the correspondence between deep neural networks and human representations," *Cognitive science*, vol. 42, no. 8, pp. 2648–2669, Nov. 2018, publisher Copyright: © 2018 Cognitive Science Society, Inc.

[6] P. Tarigopula, S. L. Fairhall, A. Bavaresco, N. Truong, and U. Hasson, "Improved prediction of behavioral and neural similarity spaces using pruned dnns," *Neural Networks*, vol. 168, pp. 89–104, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608023004690

[7] N. Truong, D. Pesenti, and U. Hasson, "Explaining human comparisons using alignment-importance heatmaps," *Computational Brain & Behavior*, vol. 8, no. 3, pp. 421–441, Sep 2025. [Online]. Available: https://doi.org/10.1007/s42113-025-00235-x

[8] N. Flechas Manrique, W. Bao, A. Herbelot, and U. Hasson, "Enhancing interpretability using human similarity judgements to prune word embeddings," in *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, Y. Belinkov, S. Hao, J. Jumelet, N. Kim, A. McCarthy, and H. Mohebbi, Eds. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 169–179. [Online]. Available: https://aclanthology.org/2023.blackboxnlp-1.13/

[9] W. Bao and U. Hasson, "Identifying and interpreting non-aligned human conceptual representations using language modeling," 2024. [Online]. Available: https://arxiv.org/abs/2403.06204

[10] O. Khattab, A. Singhvi, P. Maheshwari, Z. Zhang, K. Santhanam, S. Vardhamanan, S. Haq, A. Sharma, T. T. Joshi, H. Moazam, H. Miller, M. Zaharia, and C. Potts, "Dspy: Compiling declarative language model calls into self-improving pipelines," 2023. [Online]. Available: https://arxiv.org/abs/2310.03714

[11] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," 2019.

[12] N. Kriegeskorte, M. Mur, and P. A. Bandettini, "Representational similarity analysis - connecting the branches of systems neuroscience," *Frontiers in Systems Neuroscience*, vol. Volume 2 - 2008, 2008. [Online]. Available: https://www.frontiersin.org/journals/systems-neuroscience/articles/10.3389/neuro.06.004.2008

[13] Z. Wu, A. Arora, A. Geiger, Z. Wang, J. Huang, D. Jurafsky, C. D. Manning, and C. Potts, "Axbench: Steering LLMs? even simple baselines outperform sparse autoencoders," in *Forty-second International Conference on Machine Learning*, 2025. [Online]. Available: https://openreview.net/forum?id=K2CckZjNy0

APPENDIX A
CONCEPT DEFINITION GENERATION DETAILS

To generate diverse and high-quality sentences for each concept, we utilized the DSPy framework [10] with the Qwen3-30B-A3B-Instruct model. We employed a Chain-of-Thought (CoT) prompting strategy to encourage reasoning and diversity in the outputs.

Two distinct signatures were defined to generate two types of positive examples: *Facts* and *Stories*. The signatures provided to the model were as follows:

- **GenerateFacts**: "Generate 10 diverse, factual sentences about the given concept. Output exactly 10 sentences, one per line, with no numbering or bullets. Refer to the concept only by its name without specific classes, types, or names. Make sure the sentences are diverse and do not repeat."
- **GenerateStories**: "Generate 10 diverse, story-based sentences about the given concept. Output exactly 10 sentences, one per line, with no numbering or bullets. Each sentence should be a short story about the concept. Refer to the concept only by its name without specific classes, types, or names."

For each concept, we provided the concept name, its indefinite article ("a" or "an"), and its WordNet definition as inputs.

A. Generation Hyperparameters and Diversity Mechanisms

Beyond the explicit diversity instruction in the prompt, several additional mechanisms were employed to maximise output variety:

- **Sampling temperature.** The language model was configured with a temperature of 0.9, promoting high sampling randomness across batches.
- **Per-batch random seed injection.** Each generation call included a unique `random_seed` input field set to a fresh random value (`str(random.random())`). Because this seed is part of the prompt text sent to the model, it perturbs the input for every batch and encourages different completions even when all other inputs are identical.
- **Post-generation deduplication.** All collected sentences were deduplicated with exact string matching while preserving order. In the asynchronous generation path, duplicates were additionally rejected in-loop before accumulation, and the generator retried until the target count of unique sentences was reached (up to $2\times$ the target number of attempts).

Table VI summarises the key generation parameters.

TABLE VI: Generation hyperparameters used for concept definition creation.

Parameter	**Value**
Model	Qwen3-30B-A3B-Instruct-2507
DSPy module	`dspy.ChainOfThought`
Temperature	0.9
Max tokens	4096
Positive facts per concept	200
Positive stories per concept	200
Negatives per concept	1000
Random seed (global)	1234

B. Generated Examples

Table VII shows representative generated sentences for one concept per semantic category.

TABLE VII: Representative generated facts and stories for one concept per semantic category.

Category	**Concept**	**Fact**	**Story**
Animals	Bear	Bears are large mammals known for their powerful build and thick fur.	A bear lumbered through the misty dawn, its massive paws sinking into the damp earth as it followed the scent of wild berries.
Body Parts	Arm	An arm is a human limb that extends from the shoulder to the elbow.	She raised her arm in victory after crossing the finish line, the wind rushing through her fingers as the crowd roared.
Buildings	Apartment	An apartment is a residential unit located within a multi-unit building.	She unpacked her last box and stood in the center of the apartment, suddenly realizing she had made it on her own.
Building Parts	Arch	An arch is a curved structure that provides support in both biological and architectural contexts.	An arch in the foot of a ballet dancer helped her leap across the stage without pain, even after years of rigorous training.
Clothing	Coat	A coat is a type of outer garment designed to cover the body from the shoulders down.	The little girl clutched her new coat as she stepped into the snow, its red hood flapping like a flag in the wind.
Furniture	Bed	A bed is a piece of furniture designed to provide a place for sleeping.	She slipped into the bed after a long day, the sheets still warm from the afternoon sun.
Insects	Ant	An ant is a social insect that lives in highly organized colonies.	An ant carried a crumb twice its size across a sunlit path, guided by a trail of scent.
Kitchen	Bottle	A bottle is commonly used to store beverages such as water, soda, and juice.	A curious child found a bottle washed up on the shore, its label faded but still hinting at a faraway land.
Tools	Chisel	A chisel is a hand tool with a flat steel blade that has a sharp cutting edge.	A sculptor carefully guided the chisel through marble, shaping the first curve of a new goddess.
Vegetables	Carrot	A carrot is a root vegetable known for its long, conical shape and vibrant orange color.	In a dream, a boy wandered through a field where every carrot glowed with its own soft light.
Vehicles	Airplane	An airplane is an aircraft designed for sustained flight using fixed wings.	An airplane soared through the storm clouds, its lights blinking like distant stars in the dark sky.
Other	Bell	A bell is a hollow metal device that produces a ringing sound when struck.	A lone bell rang at midnight, summoning villagers to the forgotten village square where time had stopped.

APPENDIX B
REDUNDANCY ANALYSIS DETAILS

Figure 4 shows the per-concept redundancy rates at redundancy threshold $r_{\min} = 0.9$. Most concepts exhibit zero or near-zero redundancy, with *horse* showing the highest rate (0.8%).

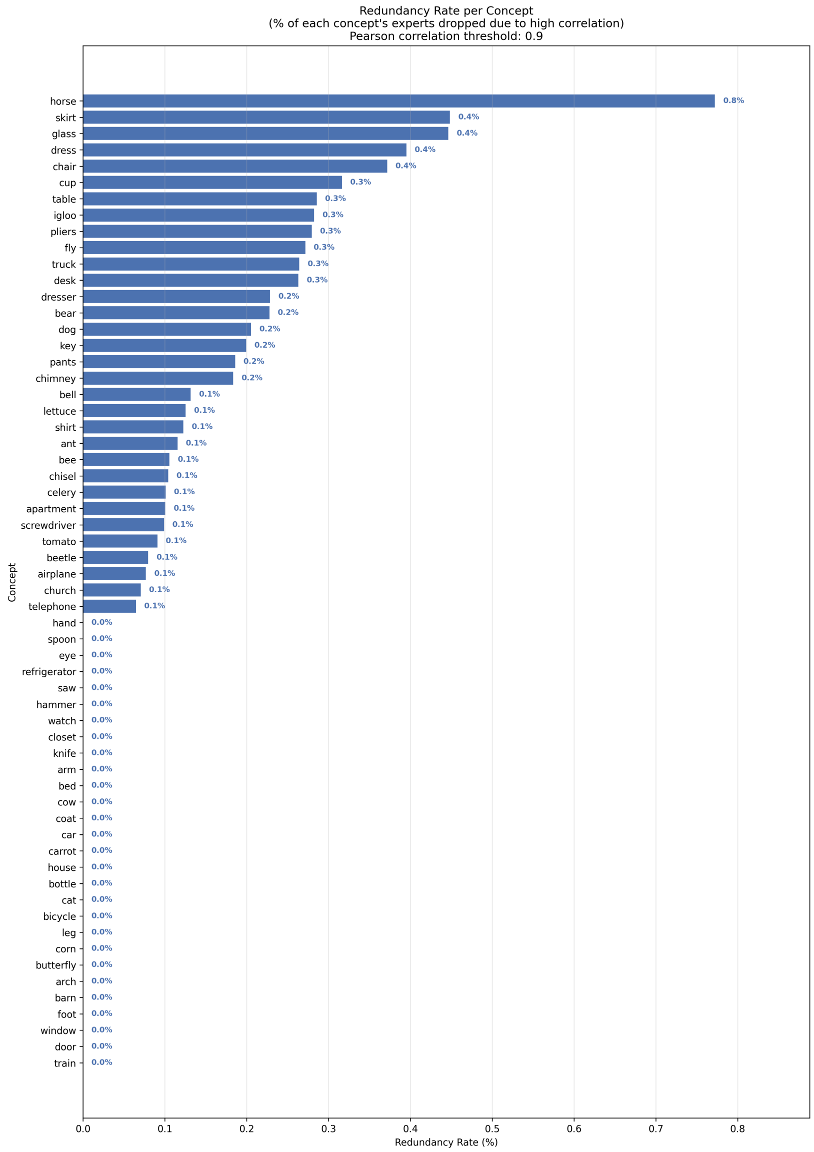


Fig. 4: Redundancy rate (%) per concept at $r_{\min} = 0.9$. Concepts sorted by redundancy rate.

Figure 5 shows the same analysis at a more permissive threshold ($r_{\min} = 0.8$), where redundancy rates increase slightly but remain below 3% for all concepts.

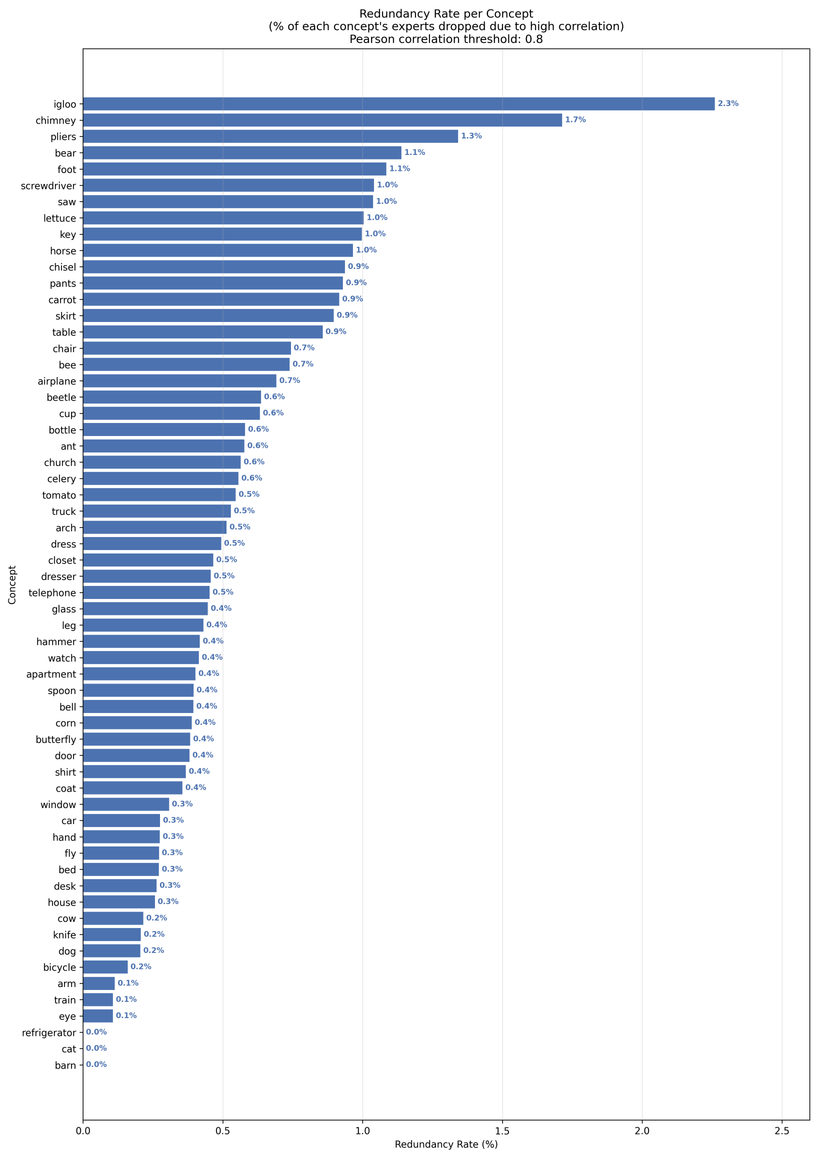


Fig. 5: Redundancy rate (%) per concept at $r_{\min} = 0.8$.

APPENDIX C
ADDITIONAL VISUALIZATIONS

A. Concept Similarity Visualizations

Figure 6 shows the full 60×60 Jaccard similarity heatmap between expert neuron sets at AP $\geq$ 0.5.

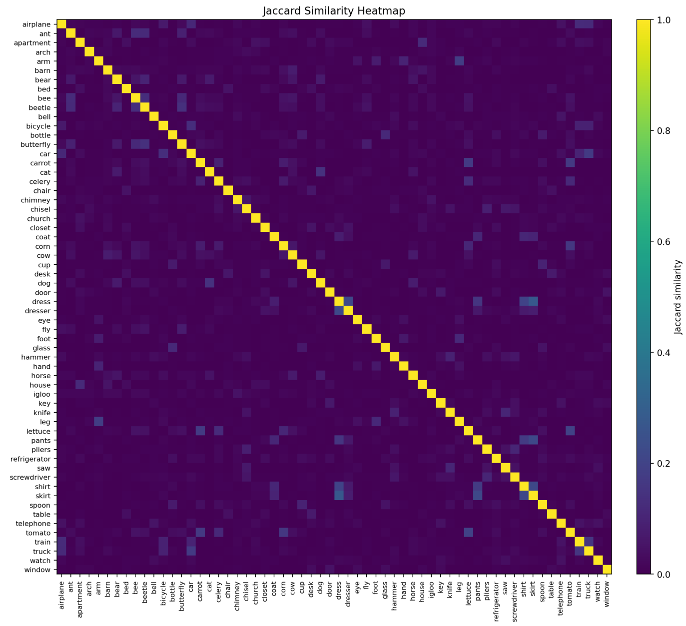


Fig. 6: Jaccard similarity heatmap between expert neuron sets for all 60 concepts.

Figure 7 shows the concept similarity graph with a lower edge threshold (5% Jaccard similarity), revealing additional cross-category connections not visible in the main text figure (10% threshold). While within-category similarities remain dominant, this visualization reveals weaker but systematic cross-category relationships.

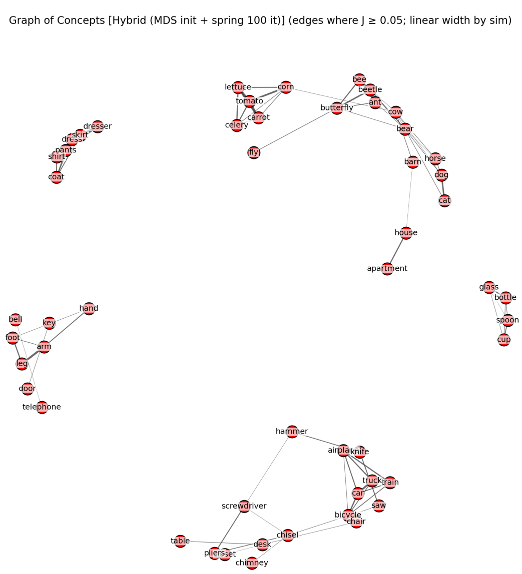


Fig. 7: Concept similarity graph with edge threshold at 5% Jaccard similarity. Additional cross-category connections become visible, though within-category clustering remains the dominant pattern.

B. RSA Sensitivity Analysis

Tables VIII present the full RSA statistics for all model components not shown in the main text.

TABLE VIII: RSA Statistics for Additional Layers

(a) Attention (`attn.c_attn`)

Cond	Sig	FDR	Mean d	Max d	Exp (SD)	Dense (SD)
0.5(U0.8)	4	0	+0.026	+1.019	0.018(.02)	0.018(.02)
0.5	5	0	+0.020	+1.012	0.018(.02)	0.018(.02)
0.5(U0.9)	5	0	+0.018	+1.011	0.018(.02)	0.018(.02)
0.6(U0.8)	2	0	-0.068	+1.000	0.017(.02)	0.018(.02)
0.6(U0.9)	2	0	-0.079	+0.955	0.017(.02)	0.018(.02)
0.6	2	0	-0.085	+0.959	0.017(.02)	0.018(.02)
0.9	9	0	-0.230	+0.464	0.005(.01)	0.018(.02)
0.8	12	0	-0.307	+1.017	0.010(.02)	0.018(.02)
0.7	15	0	-0.353	+0.740	0.011(.02)	0.018(.02)

(b) MLP (`mlp.c_fc`)

Cond	Sig	FDR	Mean d	Max d	Exp (SD)	Dense (SD)
0.5(U0.8)	19	4	+0.443	+2.106	0.017(.02)	0.013(.02)
0.5(U0.9)	20	2	+0.439	+1.938	0.017(.02)	0.013(.02)
0.5	20	2	+0.437	+1.918	0.017(.02)	0.013(.02)
0.6	20	4	+0.397	+2.012	0.018(.02)	0.013(.02)
0.6(U0.9)	20	4	+0.397	+2.006	0.018(.02)	0.013(.02)
0.6(U0.8)	21	5	+0.388	+2.161	0.018(.02)	0.013(.02)
0.7	13	2	+0.157	+2.124	0.015(.02)	0.013(.02)
0.9	10	0	-0.183	+0.707	0.005(.01)	0.013(.02)
0.8	12	4	-0.207	+0.976	0.007(.02)	0.013(.02)

C. Unique Expert Comparison

As a robustness check, we computed concept similarity using only unique experts (after redundancy filtering). Figures 8 and 9 show the concept graphs at redundancy thresholds $r_{\min} = 0.9$ and $r_{\min} = 0.8$, respectively. The top similarity pairs remained nearly identical across conditions (differences $< 0.5\%$), indicating that the conceptual organization is robust to redundancy filtering.

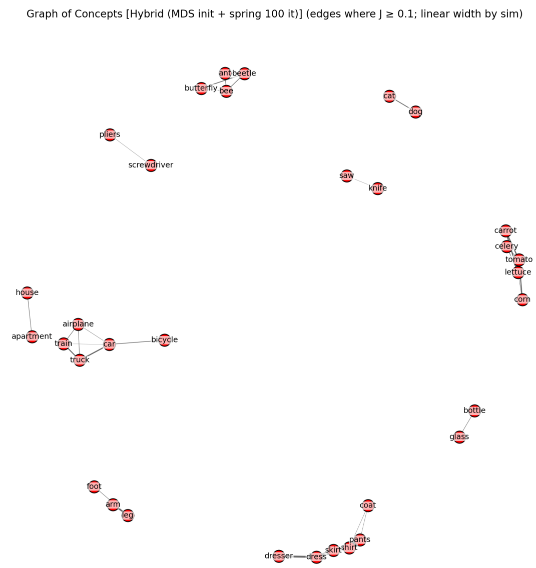


Fig. 8: Concept similarity graph using unique experts ($r_{\min} = 0.9$).

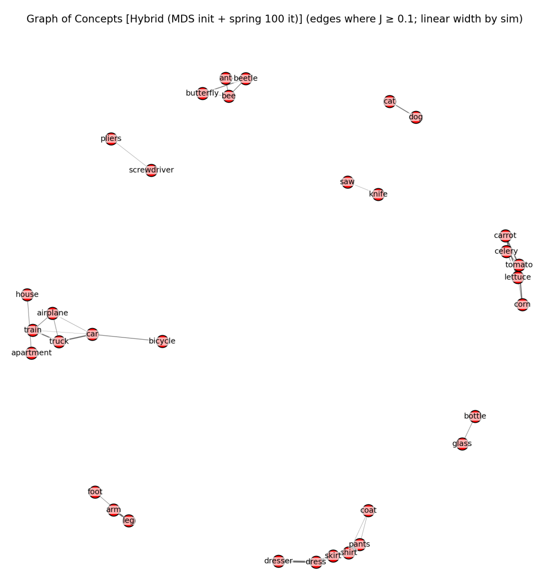


Fig. 9: Concept similarity graph using unique experts ($r_{\min} = 0.8$).

D. *Unique Expert Filtering Algorithm*

Algorithm 1 describes the procedure for filtering redundant expert neurons.

Algorithm 1 Unique Expert Filtering

Require: Expert set $\mathcal{E}_l(c)$, redundancy threshold $r_{\min}$
Ensure: Unique expert set $\mathcal{E}_l^{\text{unique}}(c)$

```
Compute pairwise Pearson correlations ρ_mn for all m, n ∈ E_l(c)
Build graph G = (V, E) where V = E_l(c), E = {(m, n) : ρ_mn > r_min}
Find connected components {G_1, ..., G_k} in G
E_l^unique(c) ← ∅
for each component G_i do
    m* ← argmax_{m∈G_i} AP(m, c)
    E_l^unique(c) ← E_l^unique(c) ∪ {m*}
end for
return E_l^unique(c)
```

Compute pairwise Pearson correlations ρ_{mn} for all $m, n \in \mathcal{E}_l(c)$
Build graph $G = (V, E)$ where $V = \mathcal{E}_l(c)$, $E = \{(m, n) : \rho_{mn} > r_{\min}\}$
Find connected components $\{G_1, \ldots, G_k\}$ in G
$\mathcal{E}_l^{\text{unique}}(c) \leftarrow \emptyset$
for each component G_i **do**
    $m^* \leftarrow \arg\max_{m \in G_i} \text{AP}(m, c)$
    $\mathcal{E}_l^{\text{unique}}(c) \leftarrow \mathcal{E}_l^{\text{unique}}(c) \cup \{m^*\}$
end for
return $\mathcal{E}_l^{\text{unique}}(c)$

APPENDIX D
PRELIMINARY STEERING VALIDATION

Our main contribution is the RSA-based brain alignment analysis presented in Section IV-C. We include preliminary steering experiments as complementary causal validation of the expert identification method. Following Suau et al. [3] and Fedzechkina et al. [2], who used steering to confirm that AP-identified experts causally control concept generation, we verify that the same holds for the experts we identified for the Mitchell et al. vocabulary. This provides evidence that the neurons showing brain alignment genuinely encode concept-level information, rather than reflecting incidental activation patterns selected by the AP threshold. We employ vector addition steering inspired by recent work on concept vectors [13].

A. Methodology

We extract concept vectors using the DiffMean method, which computes the direction separating positive and negative sample means:

$$\mathbf{v}_c = \frac{\mu_{\text{pos}} - \mu_{\text{neg}}}{||\mu_{\text{pos}} - \mu_{\text{neg}}||}$$

During generation, we add the scaled concept vector to hidden states:

$$\mathbf{h}' = \mathbf{h} + \alpha \cdot \text{max\_act} \cdot \mathbf{v}_c$$

where α is a steering coefficient and max_act is the maximum activation observed for the concept. We generate text using valid neutral prompts (e.g., "Once upon a time", "The story begins with") to ensure that any concept-specific content arises solely from the steering vector rather than the input context.

B. Preliminary Results

We tested steering on two layer type combinations: `attn.c_proj` with `mlp.c_fc` (Figure 10), and `attn.c_attn` with `mlp.c_fc` (Figure 11). For each configuration, we evaluate three vector types: *global* (all neurons), *masked* (expert neurons only), and *killed* (non-expert neurons only).

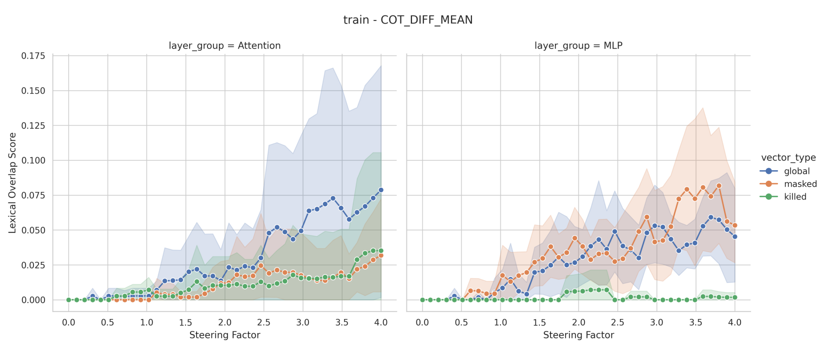


Fig. 10: Steering validation for concept "train" using DiffMean vectors from `attn.c_proj` and `mlp.c_fc` layers. Lexical overlap score measures how well the generated text matches the target concept. Higher steering coefficients increase concept presence.

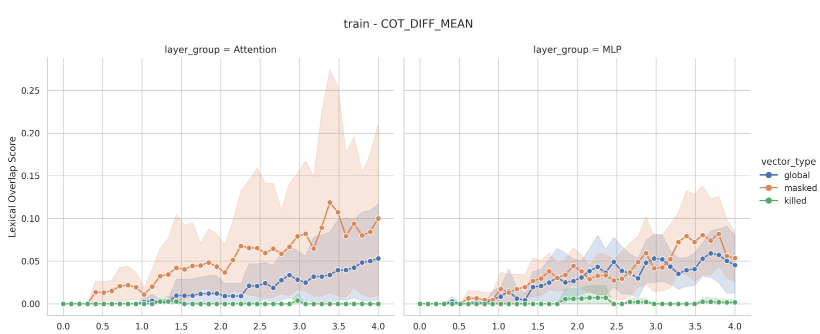


Fig. 11: Steering validation for concept "train" using DiffMean vectors from `attn.c_attn` and `mlp.c_fc` layers.

a) Example Generations: At coefficient $\alpha = 3.0$ for the "train" concept, the model generates text containing train-related words:

> *"Once upon a time, the station was running as planned, and had the station stop over at the station stop at Pearson Station. The train stopped at the station at Pearson for a few minutes, then stopped on the train..."*
>
> *"The story begins with a strange time from the train stop to a massive train station in Berlin that trains stop at the station stops. The train station trains run on the same tracks as at stations..."*

C. Concept Vector Geometry

To understand the structure of the extracted concept vectors, we visualize their pairwise cosine similarities as heatmaps. Figure 12 shows heatmaps for `attn.c_proj` at layer 6, comparing global vectors (all neurons), masked vectors (expert neurons only), and killed vectors (non-expert neurons). Figure 13 shows the same comparison for `mlp.c_fc`.

a) Limitations: These results are preliminary. While we clearly observe the target concept appearing in the text, the model often produces repetitive or nonsensical outputs. This repetitive behavior, likely due to the limited capacity of GPT-2 Small, makes it difficult to quantify the precise strength of the steering effect. Future work should validate this approach across a wider range of concepts and use more robust evaluation metrics to confirm the findings.

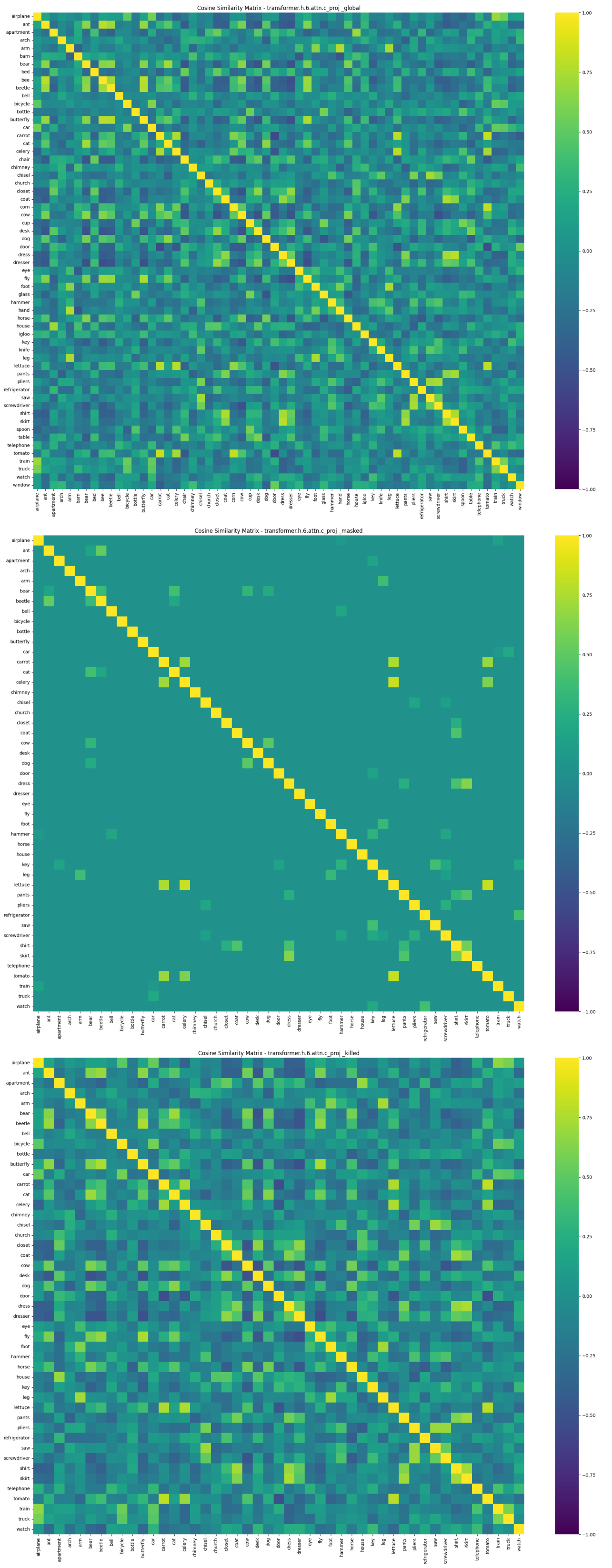


Fig. 12: Cosine similarity heatmaps for concept vectors from `attn.c_proj` layer 6. Top: global (all neurons). Middle: masked (experts only). Bottom: killed (non-experts only). Semantic clustering is visible in all conditions.

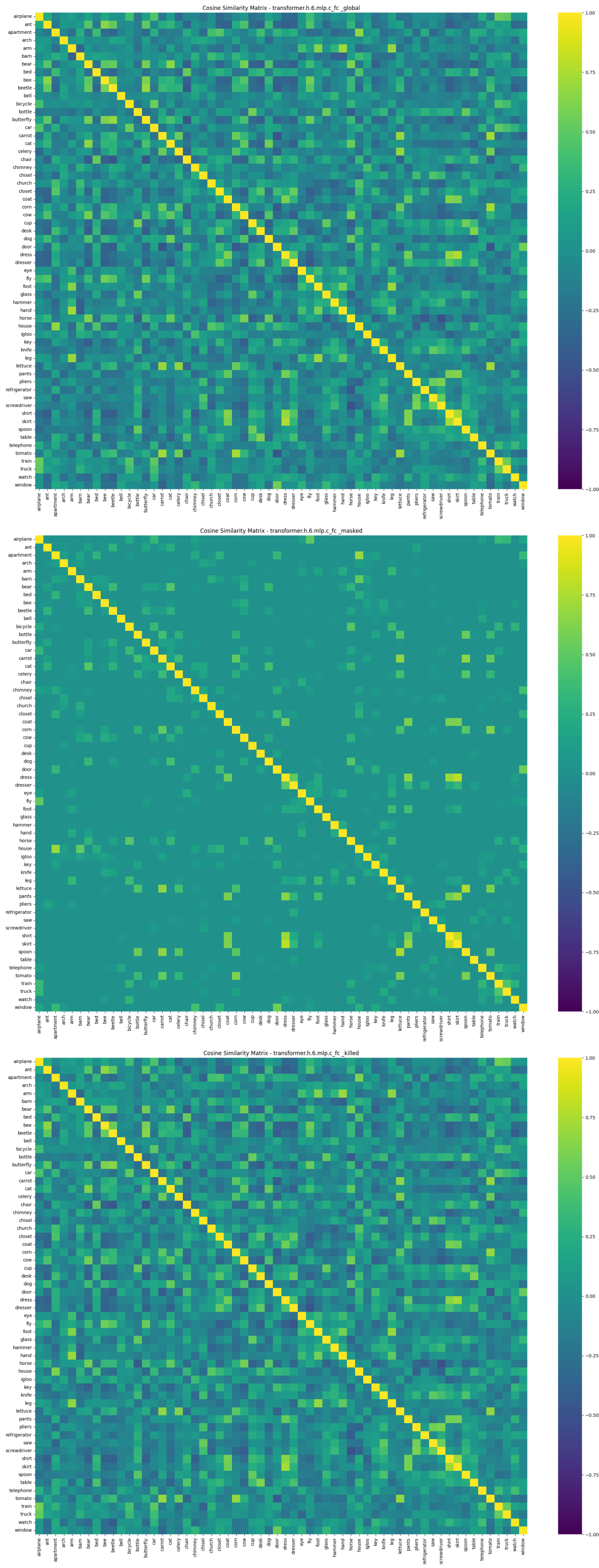


Fig. 13: Cosine similarity heatmaps for concept vectors from `mlp.c_fc` layer 6. Top: global. Middle: masked (experts only). Bottom: killed (non-experts only).